Units 1, 2, 3
Data Storage

1.
(a)

```
  01100000
+ 11100100
----------
```

1 01000100_2 ---overflow, ignoring the carry beyond 8 bits = $01000100_2 = 68_{10}$

- $01100000_2 = 96_{10}$
- Since 11100100_2 first bit is **1,** this suggests it is a **negative** number in two's complement.
  - 1's complement: 00011011
  - 2's complement: $00011100 = 28_{10}$
  - Since the original number was negative, $11100100_2 = -28_{10}$
- $96 + (-28) = 68$

Ans: 68

(b)
- Represent 01101_2 and 00111_2 in 8-bit:
  00111 = 00000111
  01101 = 00001101
- Compute the two's complement of 00001101_2:
  - 1's complement: 11110010
  - 2's complement: 11110011

```
  00000111
+ 11110011
----------
```

11111010_2

(2's complement of 11111010 is 00000110, which is 6. So 11111010 is -6)

Ans: -6

(c)
- Represent $(-100)_{10}$ in 8-bit excess:

$-100 + 128 = 28_{10} = 00011100_2$

- Represent $(-90)_{10}$ in 8-bit excess:

$-90 + 128 = 38_{10} = 00100110_2$

00011100
+ 00100110

$01000010_2 = 66_{10}$

66 − 128 = − 62

Ans: -62

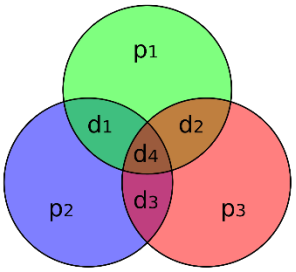
2.

steps	input	output code	new dict
1	y	7	-
2	o	4	-
3	u	6	-
4	_	8	9: you
5	s	5	-
6	e	1	-
7	e	1	-
8	_	8	10: see
9	n	3	
10	o	4	
11	_	8	11: no
12	s	-	-
13	e	-	-
14	e	10	-
15	_	8	-
16	y	-	-
17	o	-	-
18	u	9	-
19	_	8	-
20	n	-	-
21	o	11	-
22	_	8	-
23	s	-	-
24	e	-	-
25	e	10	-
26	_	8	-
27	i	2	-
28	_	8	-
29	s	-	-
30	e	-	-
31	e	10	-

Ans: 7, 4, 6, 8, 5, 1, 1, 8, 3, 4, 8, 10, 8, 9, 8, 11, 8, 10, 8, 2, 8, 10

3.

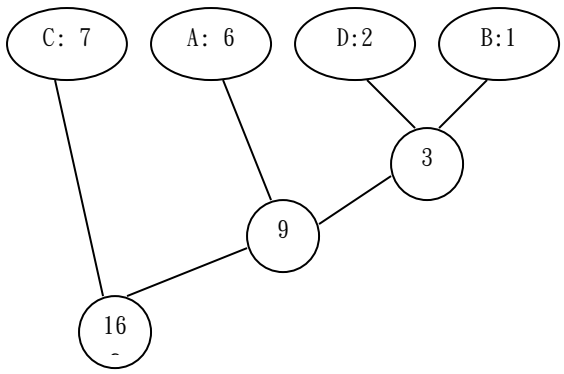
- Received message
  - $p_1 = 1, p_2 = 0, p_3 = 1$
  - $d_1 = 1, d_2 = 1, d_3 = 1, d_4 = 0$



- Calculate the check bit
  - $p_1 = d_1 \oplus d_2 \oplus d_4 = 0 \rightarrow$ different from received $p_1 = 1 \rightarrow$ error!
  - $p_2 = d_1 \oplus d_3 \oplus d_4 = 0 \rightarrow$ same as received $p_2 = 0$
  - $p_3 = d_2 \oplus d_3 \oplus d_4 = 0 \rightarrow$ same from received $p_3 = 1 \rightarrow$ error!
- Fix the error bit: $d_2 = 1 \rightarrow d_2 = 0$
- original 4-bit message = 1 0 1 0

Ans: 1010

4.



C: 1 bit, A: 2 bits, D: 3 bits, B: 3 bits

$7 \times 1 + 6 \times 2 + 2 \times 3 + 1 \times 3 = 28$

Ans: 28 bits